IMPROVING SOCIETY REGISTRATION SYSTEM: A SIMULATION-DRIVEN AI-BASED BUSINESS PROCESS ENGINEERING APPROACH

Abstract

Students look for faster, more efficient ways to do their assignments without wasting time or money in the fast-paced world of today. Processes can be redesigned using business process engineering (BPE) techniques to increase productivity, cut down on delays, and accomplish strategic objectives. In this study, we use BPE in conjunction with modeling and simulation approaches to examine a university's society registration system. We discovered through surveys and observation that the manual registration process causes errors, long lines, and missed lectures for students. We suggested an AI-supported online society registration system to address these problems, with the goals of cutting down on waiting times, minimizing human error, and improving user experience. Using AnyLogic simulation software, we created models of the current manual system, the digital system, and the AI-based system. According to the results, the AI-based method greatly increased student happiness, cut down on registration time to just one minute, and completely eliminated lines. This study demonstrates that using simulation and BPE not only addresses the fundamental issues but also offers a straightforward, effective, and resource-efficient implementation route.

Keywords: Society registration, Business Process Engineering, Simulation and Modeling, AI-based system

Introduction

BPE focuses on rethinking processes to increase efficiency and alignment with strategic goals**.** Businesses can reduce expenses, boost customer satisfaction, and obtain a competitive advantage by streamlining their operations. Better decision-making results from the risk-free testing and outcome prediction that simulation and modeling offer **[1].** Without doing experiments in the actual world, they assist organizations in understanding complicated processes and optimizing resources. Businesses can verify process modifications and forecast their effects by combining BPE with simulation, which guarantees more seamless transitions **[2].** University societies are important in providing opportunities and building network. As student's participation increasing, the demand for efficient system also increases. However the traditional system fails to provide this efficiency. Students have to miss lectures and wait in long queues for registration which mainly results from manual system which is time consuming process. After consultations and discussions with the students who have joined the society, the top issues to be rectified in the hospital are determined as follows:

Students have to wait in long lines at multiple points to request information and submit the registration form. Since staff are limited, handling each student's request individually results in delays and overcrowding. They have to collect, complete, and submit paper forms. Forms can be lost or misplaced, or contain errors requiring resubmission, adding workload and delays. It takes

a long time to manually check and approve submitted forms. If any errors are found students have to start over by filling up a new form. Students do not have easy access to information about application deadlines, eligibility requirements, and open positions, which leads to confusion and missed opportunities. Paper applications are vulnerable to loss, theft or altered without authorization, creating security and confidentiality concerns.

This study explores the development of society registration system and its implementation in order to accelerate the registration process and optimizing the long waiting queues. By developing an online application system, the students can register without hazard and in no time which will improve the overall process.

Literature Review

Business Process Engineering (BPE) focuses on analyzing and redesigning workflows and processes within an organization to improve efficiency, reduce costs, and better align with the strategic goals **[3].** BPE is critical in optimizing organizational processes, improving operational efficiency, and reducing customer waiting times, particularly in sectors like healthcare **[4].** By reengineering processes, businesses can adapt more quickly to changes and meet customer expectations effectively **[5].** A public hospital utilizes BPE techniques to improve patient flow by reducing waiting times and enhancing the overall service delivery **[3].** Simulation and modeling are techniques used to create virtual representations of systems, allowing organizations to analyze various outcomes and make informed decisions without implementing changes in the real world **[6].** These methods allow for risk-free testing and analysis of complex systems, offering predictions and insights that inform decision-making and process optimization **[7].** Simulation models also help predict the impact of changes and optimize system performance, especially in industries like healthcare and banking **[3]**. A hospital uses system simulation to optimize its emergency department's registration procedures, reducing patient wait times and improving service efficiency **[6].** Combining BPE with simulation involves analyzing and redesigning business processes using simulation tools to test changes before implementation. This allows for a more informed and data-driven decision-making process **[8].** This integration helps validate process improvements, predict the impact of changes, and ensure that the reengineered processes perform as expected without unintended consequences. This approach enhances decision-making and process optimization **[9].** A hospital combines BPE and simulation to optimize its patient registration system, simulating the effects of process changes before implementing them to ensure reduced waiting times and smoother workflows **[4].**

BPMN is a graphical notation for representing business processes, providing a standardized way to model workflows, tasks, and decision-making processes in organizations **[5].** BPMN simplifies the communication between stakeholders and helps identify inefficiencies and bottlenecks in business processes. It provides a common language for process modeling, improving understanding and decision-making across teams **[10].** A university uses BPMN to model its student registration process, enabling better resource allocation and improving service delivery during peak registration periods **[10].**

Table 4.1: Previous Research on Society Registration System

Title	Publication Year	Problem	Solution	BPE, Simulation and Modeling
Enhancing the Manufacturing Sector with Lean Integration and Business Process Reengineering : A B2Lean Methodology Approach. **[1]**	2025	Due to ineffective and antiquated processes, manufacturing organizations find it challenging to remain competitive in a fast-changing market.	The paper puts forward the B2Lean methodology, which combines elements of (BPR) and Lean Thinking to establish a hybrid framework aimed at process optimization.	BPE
Simulation of the New Indonesian Computer University Students' Registration Systems and Their Effects on the Performance of Service Systems Processes. **[11]**	2019	This research examines the ineffectiveness within UNIKOM's student registration system, focusing specifically on the use of service officers, wait times, and overall system efficiency.	Through an examination of elements such as the effectiveness of service officers and wait durations, the research proposes that refining resource distribution and simplifying registration processes could boost system performance and minimize delays.	Simulation
Bangkok Bank at KMUTNB as a Case Study for the University's Queueing System Simulation. **[3]**	2019	Due to the high demand for its services and the limited number of service channels, the bank has long wait times. This research examines the best quantity of service	Simulation is used to find bottleneck and to improve the services of bank.	Simulation
Design for Enhancing Asset Management at State University. **[16]**	2018	The processes for managing assets in state universities were manual, and ineffective. The existing system (AISM) lacked the capability to connect with other databases.	The researchers utilized (BPR) to redesign and enhance eight essential asset management procedures. They used simulations to test redesigned workflows, identifying non-value-added activities, consolidating similar tasks.	BPR, Simulation and modeling

		This resulted in duplicate manual data entry, a rise in human error, repetitive tasks, and prolonged processing times.		
Enhancing Business Processes in District Public Services through Business Process Reengineering (BPR). **[17]**	2017	The public services provided by the district are currently slow and inefficient. A lot of complaints from citizens about service time, speed, and accessibility.	The proposed solution is to apply (BPR) to the service delivery process in order to make it faster, more accessible, and more efficient.	Modeling
Research on System Simulation-Based Registration Procedure Optimization in Emergency Departments. **[6]**	2016	The registration procedure at the West China Hospital's Emergency Department (ED) was not optimized, leading to lengthy patient wait times and ineffective use of resources. In particular, the preview triage area experienced heavy congestion.	A discrete event simulation model representing the current registration process was created with Arena software. As a result of the analysis, a new registration procedure was developed that features enhanced resource allocation. As demonstrated by the simulation of the new model, average waiting time was decreased.	Simulation and Modeling
Student Industrial Internship Web Portal. **[12]**	2008	The process of managing the Student Industrial Internship Program (SIIP) at Universiti Teknologi PETRONAS (UTP) was manual, leading to inefficiencies and a high likelihood of mistakes.	Creation of the Student Industrial Internship Web Portal (SIIWP), a web-based system designed to automate the business processes of SIIP. The system, developed with PHP, MySQL, Apache, and Dreamweaver, automates tasks.	BPR
Reducing Wait Times and Customizing Patient Care with Mobile Messaging in Floating Appointments. **[13]**	2008	Ineffectiveness in the management of hospital appointments, which leads to protracted waiting periods, a considerable number of canceled appointments, staff time that is not fully utilized, and communication obstacles within healthcare service	Establishment of a mobile messaging system for floating appointments that automatically proposes suitable appointment times to patients through SMS and confirms them upon patient consent	BPR and Simulation and Modeling

		procedures.		
Is it Better to Pool or Not? The Advantages of Putting Simulation and Queuing Together . **[14]**	2002	Investigates whether pooling separate queues into a single queue is beneficial in different real-life and industrial settings.	Pooling helps to reduce waiting times and enhance system efficiency.	Simulation
Regarding the Hybrid Integration of Simulation and Queueing. **[15]**	2000	Traditional queueing theory is too mathematical and not always practical.	This paper suggests a hybrid approach that merges queueing theory with simulation in order to take advantage of the strengths of both methods.	Simulation and Modeling

Summary: The goal of Business Process Engineering (BPE) and simulation is to improve workflows, decrease human errors, reduce delays, and boost customer satisfaction through automation and technologies such as AI. BPE and simulation both assist in minimizing lengthy waiting queues and enhancing the system's efficiency. Likewise, in the Society Registration System, the conventional manual procedure results in long lines, form mistakes, delays, and limited access to critical information. Using BPE principles and simulation, we propose an automated registration system supported by AI that digitizes forms, simplifies the approval process, enhances information access, and minimizes human error. This not only boosts efficiency but also guarantees a registration process that is more secure, user-friendly, and free of errors.

METHODOLOGY

The core problem was identified at the beginning of the methodology by critically examining the university's present student society registration procedure. The procedure is extremely time-consuming and ineffective, frequently necessitating several trips to the registration office. Students must wait in lengthy lines, which frequently results in missed courses and needless delays. One significant problem is the lack of an appointment system or digital system, which makes the registration process even more difficult and aggravates students.

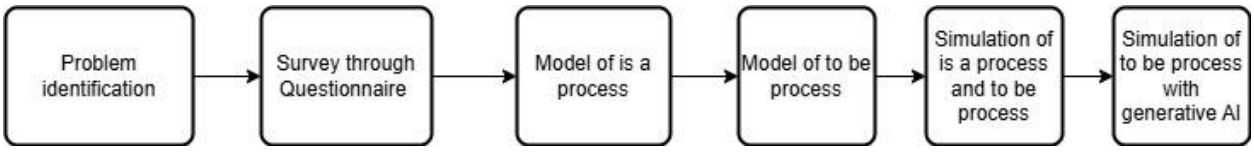


Figure 5.1: Flow Chart of the whole system

Every step of the procedure was observed during a visit to the society registration desk in order to gain a better understanding of it. Students start the process at Desk 1, where they are given basic information on the establishment of the society and the positions that are open, such as media coordinator, general secretary, and president. Application forms are also given out at this counter. Students go to Desk 2 after filling out the paperwork, where their documents are checked. Students must go back to Desk 1 to update any information that is inaccurate or missing. Students must wait several times as a result of this loop, which creates two distinct lines, one for each workstation. The number of pupils rises dramatically at times of strong demand, such societal drives, which exacerbates delays and confusion. This finding emphasizes how ineffective the current manual system is. It leads to frustration, consumes time, and raises the possibility of misunderstandings. The analysis emphasizes how urgently a digital solution that streamlines the registration process, cuts down on lines, and improves overall convenience for administrative staff and students is needed.

Our system is primarily designed to reduce the amount of time students wait during society registration and to offer them a process that is both efficient and easy to use. This research utilized primary data gathered directly from students who have participated in society registrations at least once. We collected the data using questionnaire methods from **1009** students. After the data were gathered, statistical operations were applied to analyze trends and behaviors. The results of the questionnaire revealed the following:

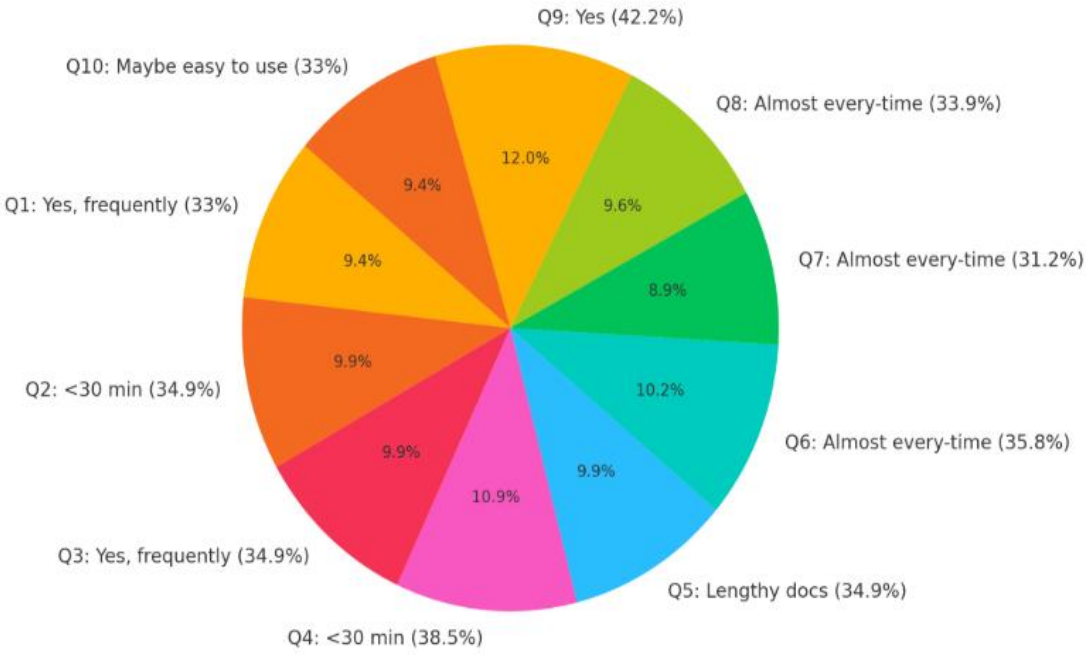


Figure 5.2: Graph of Questionnaire

Table 5.1: Results of questionnaire collected from 1009 students

Question	Answers
1. Have you ever been involved in registering a society?	Yes, frequently 33% Rarely 20.2% Occasionally 16.5% Never 30.3%
2. How long did it take for your society to get registered?	Less than 30 minutes 34.9% 30 minutes – 1 hour 30.3% 1–2 hour 15.6% More than 2 hours 19.3%
3. Have you ever experienced missing important deadlines due to unclear information?	Yes, frequently 34.9% Rarely 17.4% Occasionally 28.4% Never 19.3%
4. How long you wait in the queues for registration at society registration desk?	Less than 30 minutes 38.5% 30 minutes – 1 hour 33.9% 1–2 hour 20.2% More than 2 hours 7.3%
5. What was the biggest challenge you faced in registering a society?	Lack of clear information 31.2% Lengthy documentation process 34.9% Delayed approvals 19.3% Inefficient staff response 14.7%
6. Were you provided with clear guidelines and assistance during the registration process?	Almost every-time 35.8% Very often 21.1% Rarely 27.5% Never 15.6%
7. Have you ever observed favoritism during Society registration?	Almost every-time 31.2% Very often 26.6% Rarely 22.9% Never 19.3%
8. Did you face any difficulties in submitting the required documents physically?	Almost every-time 33.9% Very often 26.6% Rarely 25.7% Never 13.8%
9. Have you ever missed a class due to society meetings or events?	Yes 42.2% No 31.2% Rarely 13.8% Often 12.8%
10. Would an online registration system be convenient and easy to use?	No, I prefer the current manual system 8.3% Maybe it is easy to use 33% Yes, it would speed up the process 32%

	Definitely, it should replace manual system 26.6%

The data shows a significant need to improve the registration process, especially by cutting down waiting time, clarifying information, and moving toward digital solutions for greater efficiency and accessibility. After the system requirements have been collected, the next step is System Design. The suggested way to enhance the existing manual society registration process is to create an application that allows students to apply for registration. After submission, a specific appointment time will be assigned to the student digitally. They will pay a visit to a digital verification desk at that time, where their documents will be verified in a swift and effective manner.

This method removes the necessity of forming long lines and waiting in person, which leads to a considerable decrease in time expenditure and a more streamlined process. The system incorporates a token-based QR code created via the app, which the student will show at the verification point. Should the token prove valid, they will be granted permission to move forward with submitting documents and continuing their registration processing.

The aim of this process is to cut down on time as much as possible, prevent crowding, and enable students to finish their registration in the least time possible. The entire process has been illustrated in a flowchart, making the system structure easy to comprehend and implement.

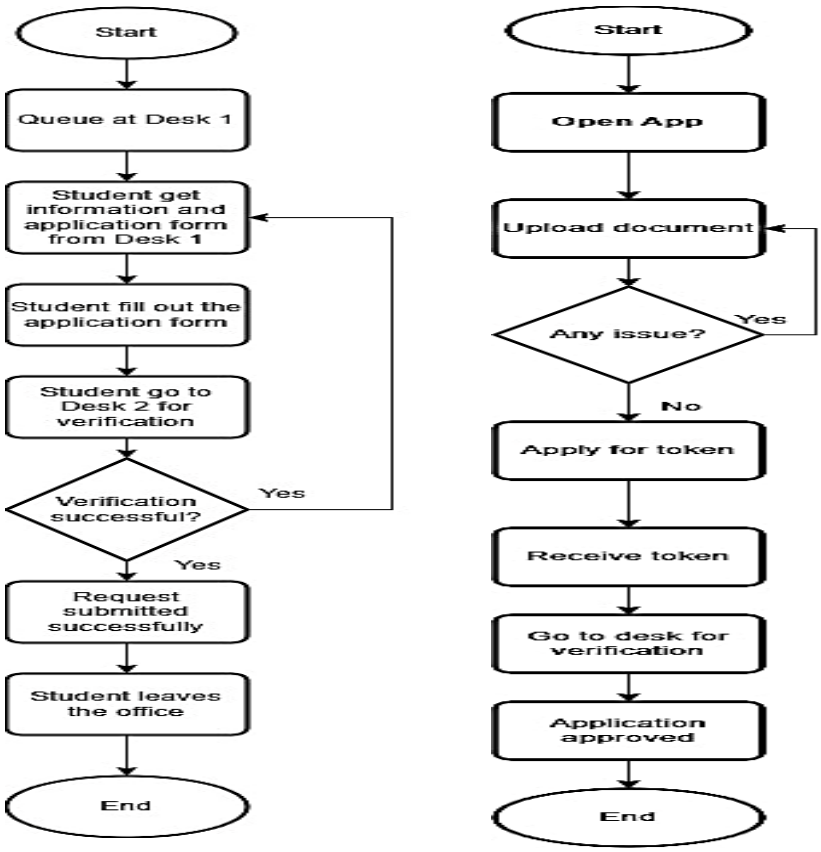


Figure 5.3: Flowchart of Is-a Process and To-be Process

The subsequent phase involves modeling the behavior of all participants and subsystems involved, such as students, staff at the registration desk, and the app itself.

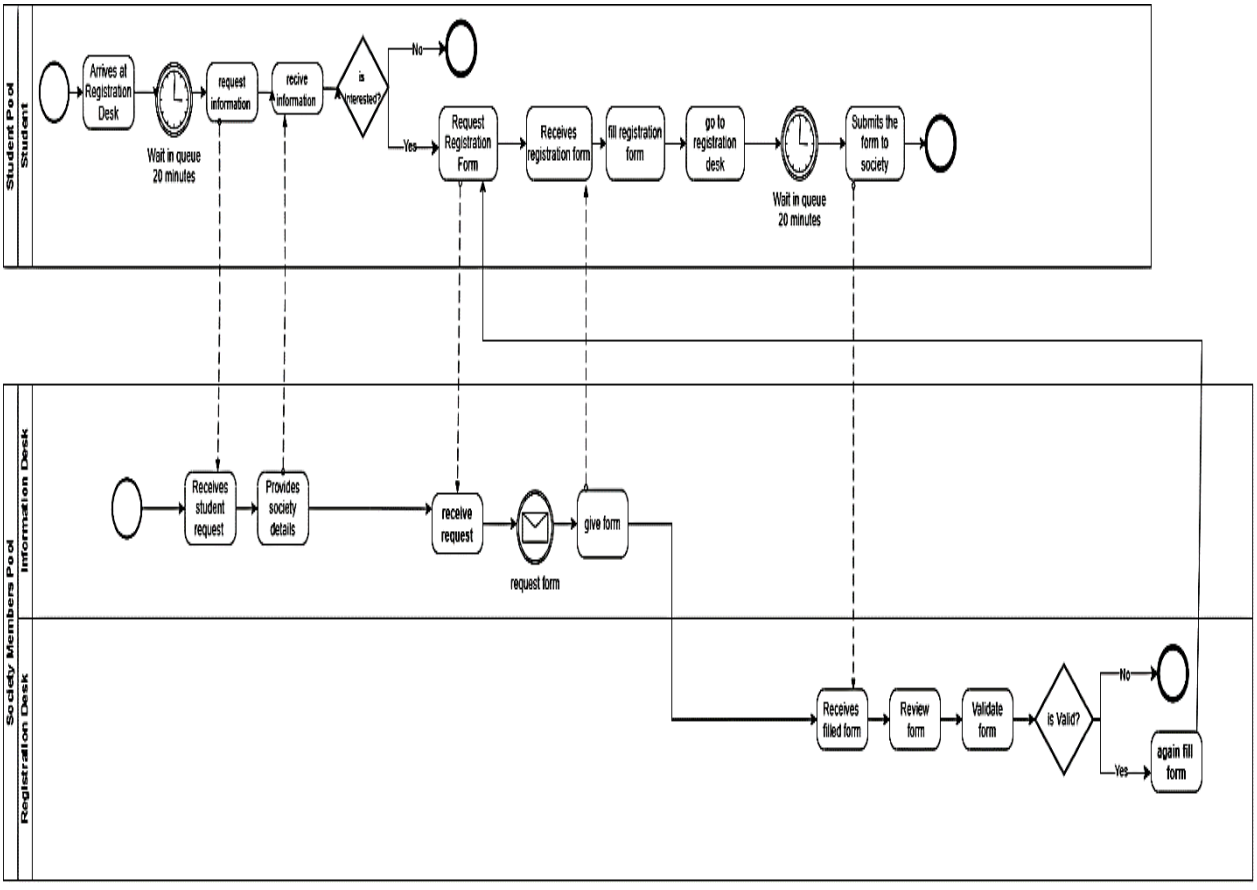


Figure 5.4: BPMN of Is-a process

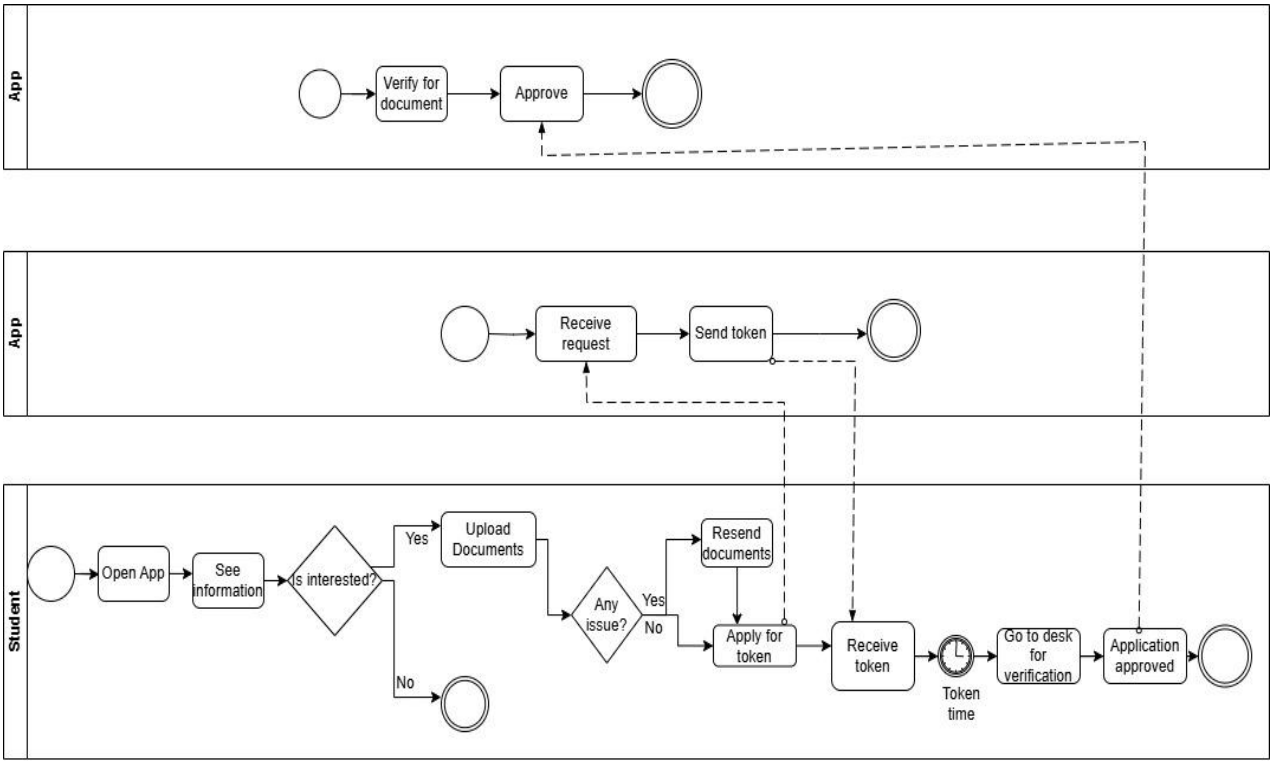


Figure 5.5: BPMN of To-b process

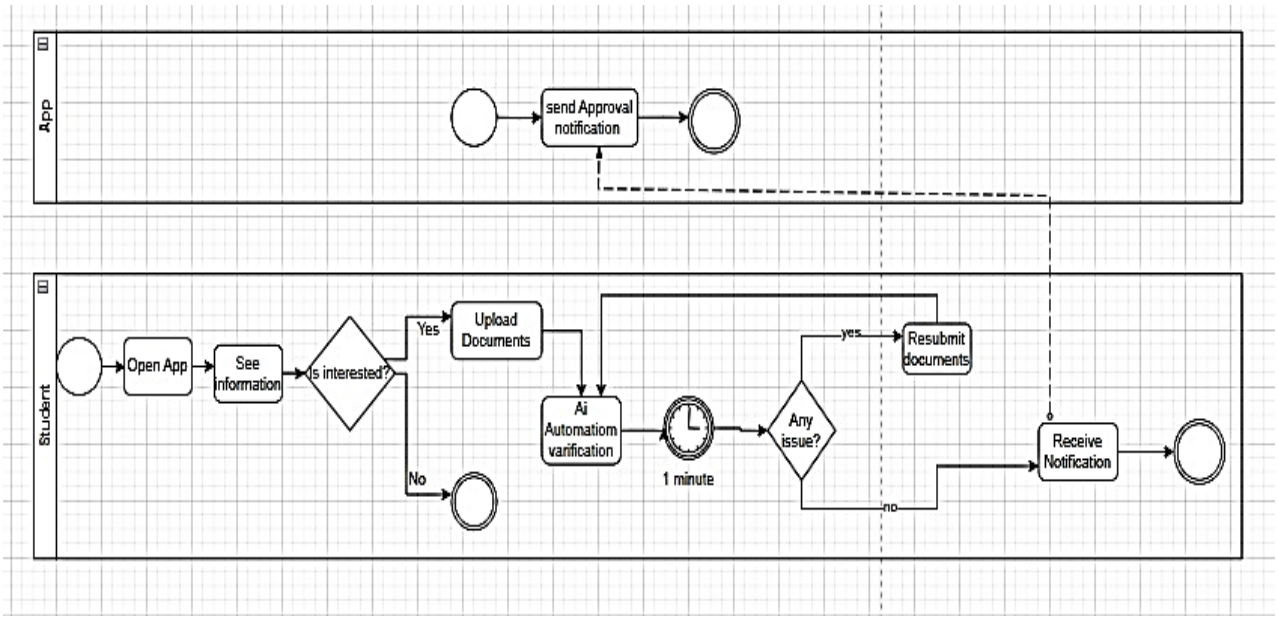


Figure 5.6: BPMN of AI generated solution

We used AnyLogic, a simulation tool, to implement this behavioral model and test the proposed solution in a virtual environment. Testing of the system verified that the new model reduces student waiting time and improves the registration process as a whole by enhancing its transparency and manageability. Moreover, the simulation aided in pinpointing and addressing possible bottlenecks as they occurred, thereby guaranteeing the efficiency of the proposed system.

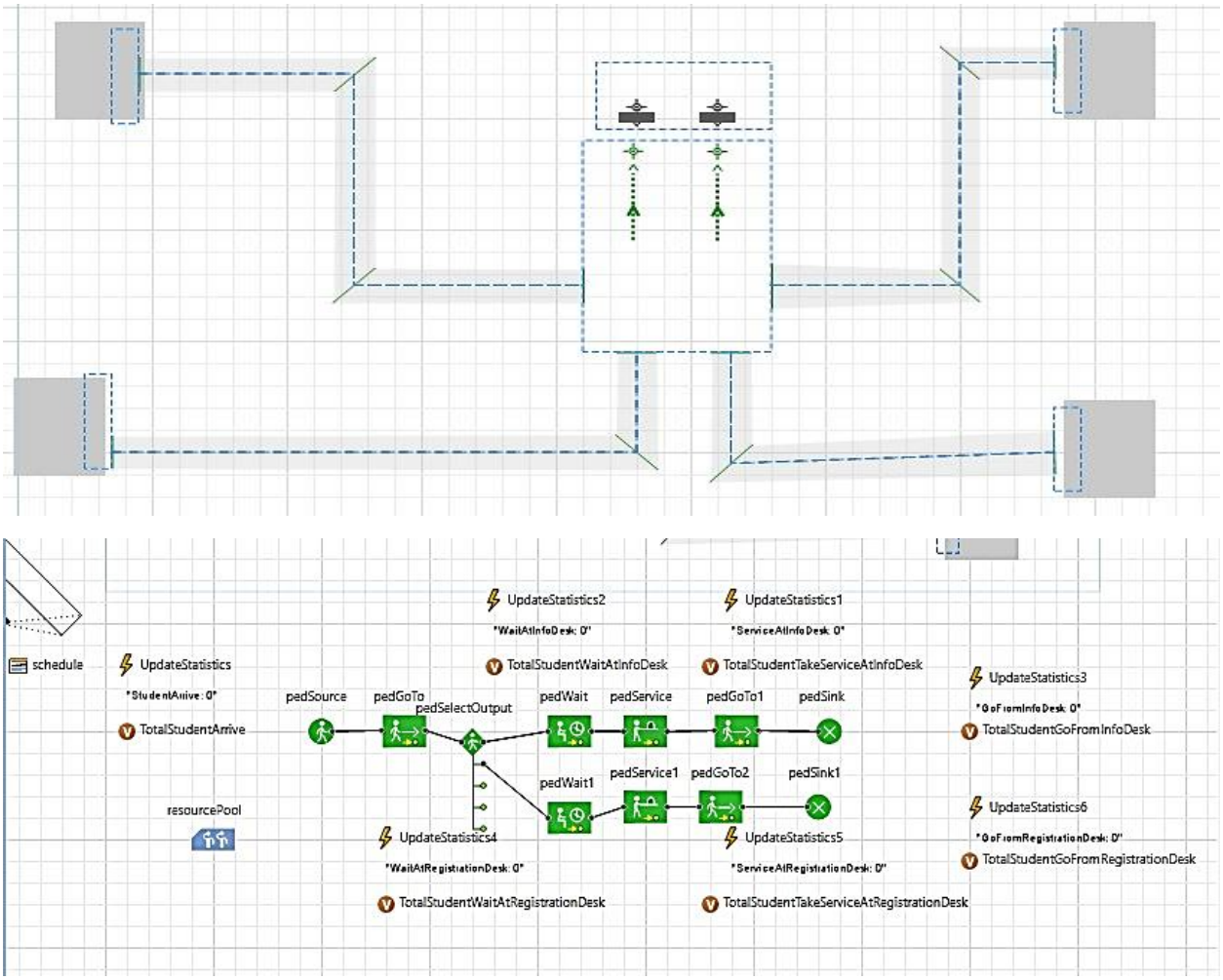


Figure 5.7: Simulation of Is-a process

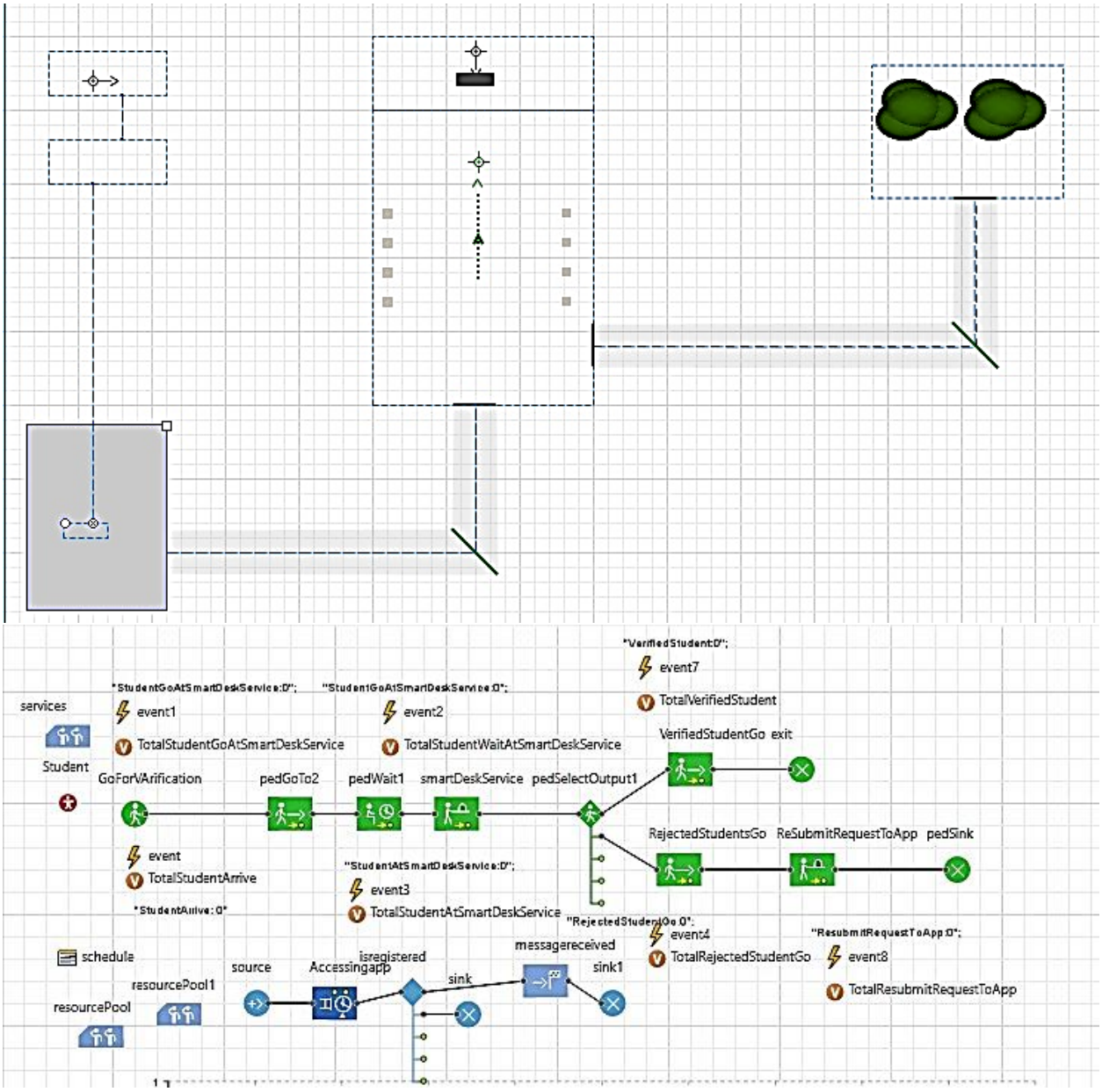


Figure 5.8: Simulation of To-be Process

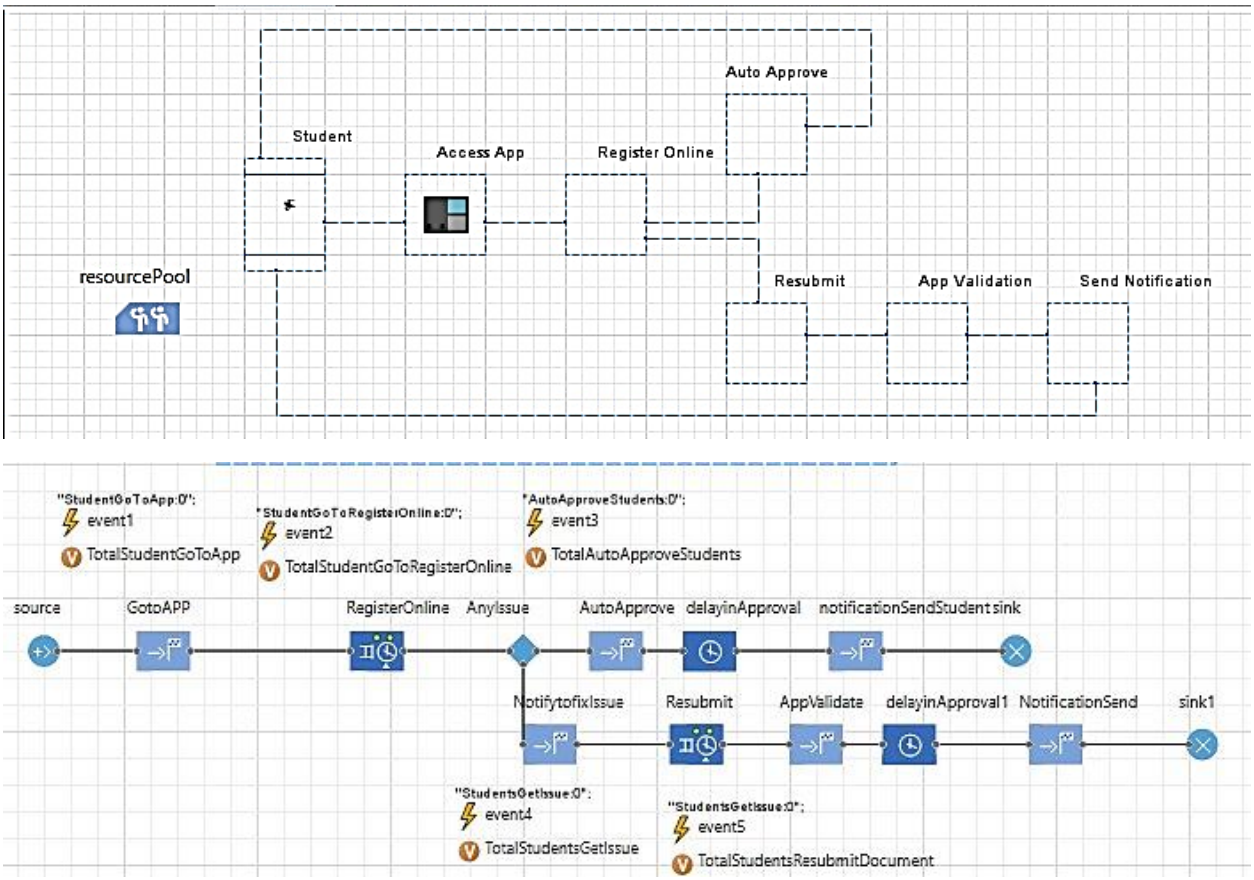


Figure 5.9: Simulation of AI generated solution model

Results and Discussion

1009 students show up for society registration, according to the IS-A process simulation model. Of these, 709 students successfully and without significant difficulties finish their registration, while 200 students are left waiting because of lengthy lines and inefficient manual processes. The entire registration process takes about 20 minutes for a student.

Table 6.1: Result of Is-a Process

Number of Students	1009
Students Registered Successfully	809
Students Waiting	200
Time Taken	20 minutes

According to the simulation model of the to-be procedure, 1009 students use the upgraded system to reach the registration area. Only 20 students are left waiting out of these, mostly as a result of sporadic system bottlenecks, while 959 students successfully finish their registration. A student's registration takes about five minutes to complete.

Table 6.2: Result of To-be Process

Number of Students	**1009**
Students Registered Successfully	989
Students Waiting	20
Time Taken	2 minutes

According to the AI-generated process simulation model, all 1009 students show up to the AI station to get help with registration. Thanks to quick processing and astute registration slot distribution, all 1009 students were able to successfully finish their registration without having to wait. Each student's complete registration process takes just one minute.

Table 6.3: Result of AI-Generated Process

Number of Students	**1009**
Students Registered Successfully	1009
Students Waiting	0
Time Taken	1 minute

The discussion comes to the conclusion that the AI-based approach is far more effective and efficient than the is-a and to-be processes. Every student is immediately registered in the AI-based system, meaning there is no waiting period and total pleasure. The AI solution is the most effective method for registering with the society because it eliminates the need to stand in line or fill out paperwork by hand.

Conclusion

We found that fewer students successfully completed their registration using the traditional manual Is-a method. Queues and waiting areas took up the majority of the time, which led to crowding, confusion, and annoyance among students. In order to address this issue, we suggested and evaluated a simulation modeling approach. The plan implemented an online registration system that allows students to register through application without having to wait in line. Students may swiftly complete supporting documentation, and get confirmation using this manner. According to the simulation results, the crowding and time waste problems were resolved by the To-Be system, particularly the AI-based system. We further compared the IS-A, To-Be, and AI-based models by analyzing their performance. The IS-A model showed the highest waiting time and lowest

registration success rate. The To-Be model improved the situation by introducing digital registration, reducing waiting times significantly. But both were surpassed by the AI-based model, which ensured the quickest and most effective registration process and totally eliminated waiting times.

To further statistically validate the variations in performance across the three models, an ANOVA test was used. The AI-based system had a definite advantage in cutting waiting times and increasing registration efficiency, according to the ANOVA test findings, which verified that there were notable differences between the models. Through simulation and statistical validation, our suggested approach thus not only resolves the issues with the manual registration system but also demonstrates that an intelligent, AI-driven system results in a very successful, effective, and student-friendly society registration procedure.

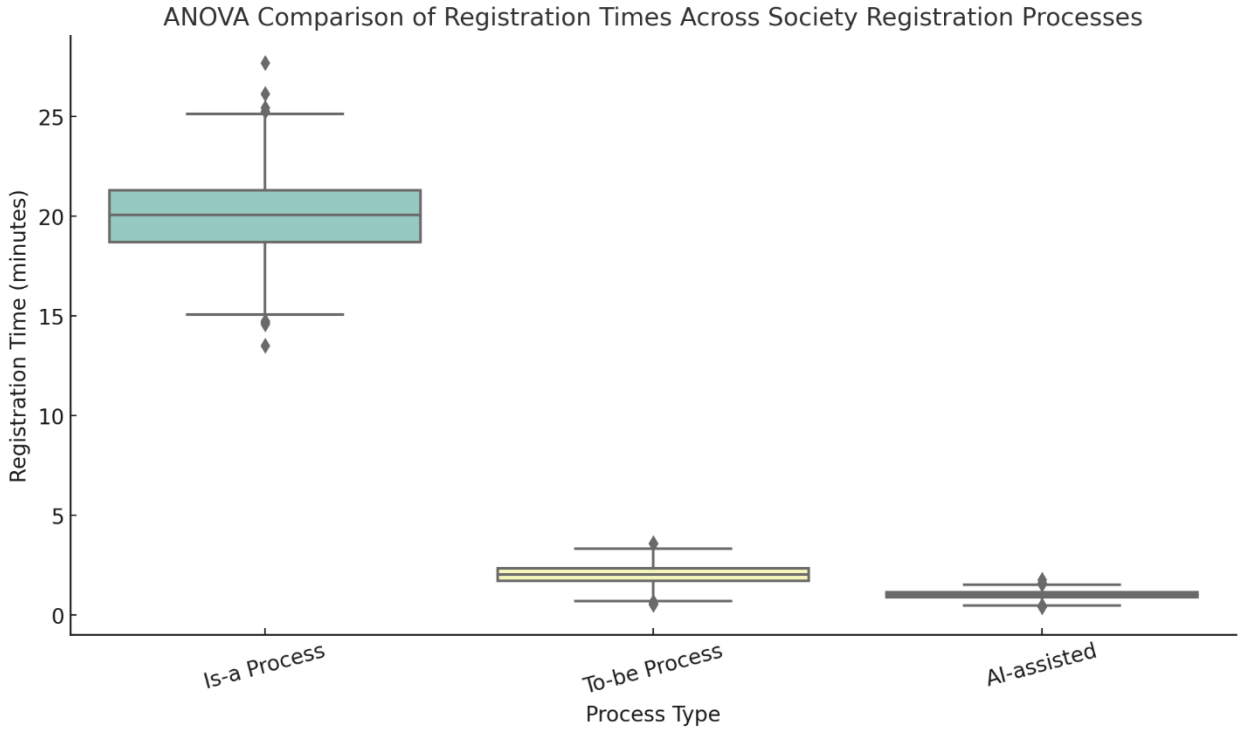


Figure 7.1: ANOVA Comparison of registration time

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